

The Impact of Forecast Characteristics on the Forecast Value for the Dispatchable Feeder

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ABSTRACT

Transforming the energy system to decentralised, renewable energy sources requires measures to balance their fluctuating nature and stabilise the energy system. One such measure is a dispatchable feeder, which combines inflexible prosumption with a flexible energy storage system. The energy storage system's management is formulated as a stochastic optimisation problem that requires energy time series forecasts as input. These forecasts can significantly influence the performance of the dispatchable feeder: the forecasts have a so-called forecast value for the dispatchable feeder, which is not directly reflected by error-based forecast quality metrics. Therefore, we analyse how the considered forecast value for the dispatchable feeder is related to the considered forecast quality and influenced by forecasts with different characteristics. Furthermore, we examine the impact of problem-specific parameters such as the data and the battery capacity. To this means, we create forecasts with different characteristics using neural networks with varying loss functions and perform the analysis using a data set with 300 buildings. The results of our analysis show that the relation between the considered forecast quality and forecast value for the dispatchable feeder is non-monotonic. Furthermore, we show that the forecast characteristics influence the forecast value differently depending on the data and the battery capacity.

CCS CONCEPTS

• **Applied computing** → **Forecasting; Decision analysis.**

KEYWORDS

forecast value, dispatchable feeder, loss function

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1 INTRODUCTION

The energy transition towards a low-carbon energy system leads to the increased integration of decentralised, renewable energy sources. Due to their fluctuating nature, measures such as dispatchable feeders are necessary to stabilise the energy system. A dispatchable feeder combines inflexible prosumption together with an intelligently managed flexible energy storage system to counteract the inflexibility inherent in the considered system [3, 11, 13]. Thereby, the management of the flexible energy storage system is formulated as an optimisation problem that requires forecasts of the prosumption as inputs.

Thus, these forecasts can have a direct impact on the performance of the dispatchable feeder, i.e. they have a so called forecast value for the dispatchable feeder. This forecast value can be measured by value metrics, which incorporate the optimisation problem's point of view by evaluating its performance. For example, in the scenario of the dispatchable feeder, the forecast value can be its economic costs. However, this forecast value is not reflected in error-based forecast metrics –such as mean absolute error (MAE) or mean squared error (MSE)– which are typically used to analyse the quality of the forecast.

This contrast between the forecast value and forecast quality is noted in numerous papers [4–6, 9]. More precisely, [6, 9] show for their particular optimisation problem that the forecast value and the forecast quality are not directly related. A more extensive examination is undertaken by [4, 5], who demonstrate for other optimisation problems that the performance of the optimisation problem and the considered forecast value is not only influenced by the forecast error but also by problem-specific parameters. Thereby, [4] includes the characteristics of the forecast to analyse the forecast value depending on the system viewpoint. However, to the best of the authors' knowledge, no previous work analyses how the forecast value for a dispatchable feeder is influenced by the forecast characteristics and problem-specific parameters.

Therefore, the contribution of the present paper is twofold. First, we show for the dispatchable feeder that the relation between the considered forecast quality and forecast value is non-monotonic and that the characteristic of the forecast impact the forecast value.

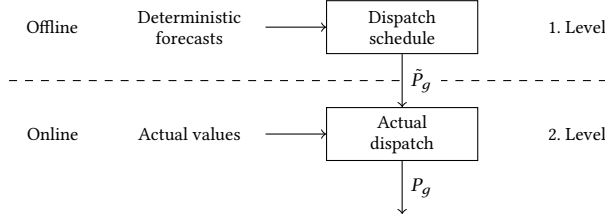


Figure 1: The two levels of the optimization problem: the offline dispatch schedule (DS) and the online actual dispatch.

Second, we show that this influence depends not only on the optimisation problem but also on problem-specific parameters. More specifically, we show the influence of different forecast characteristics on the optimisation problem's costs depending on the data and the battery capacity.

The remainder of the paper is structured as follows. Section 2.1 describes the optimisation problem of the dispatchable feeder. Section 2.2 presents the considered forecast evaluation metrics. Section 3.1 describes the experimental setup for the study and Section 3.2 presents its results. Section 4 discusses the results, while Section 5 summarises the paper and provides an outlook on further research.

2 DISPATCHABLE FEEDER

The dispatchable feeder comprises two main system components [3, 11]: an inflexible, volatile component and a flexible but power- and energy-constrained component. The primary objective of the dispatchable feeder is to intelligently manage the flexible component in order to offset the inherent system inflexibility. This management of the flexible component is achieved through a two-level non-linear optimization problem.

In this paper, we model residential buildings with rooftop PV panels and a battery as a dispatchable feeder. Thus, the inflexible component consists of the building's prosumption, which we define as domestic load minus PV power generation, and the flexible, but power- and energy-constrained component is the battery. Furthermore, the system operates discretely with time intervals indexed by $k \in \mathbb{N}$ and duration of $\Delta t \in \mathbb{R}$. Thereby, we focus on active power only. More precisely, the flexible battery is modeled via its power input $P_s(k) \in [\underline{P}_s, \bar{P}_s]$ and state of energy $E_s(k) \in [\underline{E}_s, \bar{E}_s]$ with lower and upper bounds $\underline{P}_s, \bar{P}_s \in \mathbb{R}$ and $\underline{E}_s, \bar{E}_s \in \mathbb{R}_{\geq 0}$. The state of energy evolves according to

$$E_s(k+1) = E_s(k) + \Delta t \cdot (P_s(k) - \mu P_s^+(k) + \mu P_s^-(k)) \quad (1)$$

with loss coefficient $0 \leq \mu \leq 1$ and positive and negative directions of the battery's power input $P_s^+(k) \geq 0$ and $P_s^-(k) \leq 0$. We then obtain the power exchange between the residential building and the grid by summing the battery's power input and the building's prosumption.

In the following, we first introduce the optimisation problem. Afterwards, we present the utilised forecast evaluation metrics.

2.1 Optimisation Problem of the Dispatchable Feeder

In this section, we further specify the two-level optimisation problem used to manage the battery, see Figure 1.

The first level determines offline the dispatch schedule (DS) $\tilde{P}_g(k) \in \mathbb{R}$ with respect to costs for the following day, taking into account deterministic forecasts of the prosumption $\hat{P}_l(k) \in \mathbb{R}$. Thereby, the cost function takes into account both self-consumption and peak shaving

$$C_{DS}(\tilde{P}_g^+(k), \tilde{P}_g^-(k)) = c_{\text{quadratic}}^+ \cdot (\tilde{P}_g^+(k))^2 + c_{\text{linear}}^+ \cdot \tilde{P}_g^+(k) + c_{\text{quadratic}}^- \cdot (\tilde{P}_g^-(k))^2 + c_{\text{linear}}^- \cdot \tilde{P}_g^-(k), \quad (2)$$

with positive and negative directions of the DS $\tilde{P}_g^+(k) \geq 0$ and $\tilde{P}_g^-(k) \leq 0$ and cost coefficients $c_{\text{quadratic}}^+, c_{\text{linear}}^+, c_{\text{quadratic}}^-, c_{\text{linear}}^- \in \mathbb{R}_{\geq 0}$. The first level optimisation problem is then formulated as

$$\begin{aligned} \min_{\{X\}_{\mathcal{K}}} \quad & \sum_{k \in \mathcal{K}} C_{DS}(\tilde{P}_g^+(k), \tilde{P}_g^-(k)) \\ \text{s.t. for all } k \in \mathcal{K} \quad & (1) \\ & E_s(k_0) = E_s^0 \\ & \tilde{P}_g(k) = P_s(k) + \hat{P}_l(k) \\ & \tilde{P}_g(k) = \tilde{P}_g^+(k) + \tilde{P}_g^-(k) \\ & \tilde{P}_g^+(k) \geq 0 \\ & \tilde{P}_g^-(k) \leq 0 \\ & P_s(k) = P_s^+(k) + P_s^-(k) \\ & P_s^+(k) \geq 0 \\ & P_s^-(k) \leq 0 \\ & 0 = P_s^+(k) \cdot P_s^-(k) \\ & \underline{P}_s \leq P_s(k) \leq \bar{P}_s \\ & \underline{E}_s \leq E_s(k) \leq \bar{E}_s \end{aligned} \quad (3)$$

with discrete scheduling horizon $\mathcal{K}$, decision vector $X(k) = (\tilde{P}_g(k), \tilde{P}_g^+(k), \tilde{P}_g^-(k), E_s(k+1), P_s(k), P_s^+(k), P_s^-(k))^T$, and parameters $\hat{P}_l(k), E_s^0, \underline{P}_s, \bar{P}_s, \underline{E}_s, \bar{E}_s$. Note, it is necessary to know or estimate the state of energy at the beginning of scheduling $k_0 \in \mathbb{N}$.

In the online second level, the primary objective is to minimise the deviation of the DS $\Delta P_g(k) \in \mathbb{R}$ taking into account the actual prosumption $P_l(k) \in \mathbb{R}$. Thereby, the technical constraints of the battery must be considered. The optimization problem for the

second level can be formulated as

$$\begin{aligned}
& \min_{X(k)} (\Delta P_g(k))^2 \\
& (1) \\
& E_s(k) = E_s^k \\
& P_g(k) = P_s(k) + P_l(k) \\
& P_g(k) = \tilde{P}_g(k) + \Delta P_g(k) \\
& P_s(k) = P_s^+(k) + P_s^-(k) \\
& P_s^+(k) \geq 0 \\
& P_s^-(k) \leq 0 \\
& 0 = P_s^+(k) \cdot P_s^-(k) \\
& \underline{P}_s \leq P_s(k) \leq \bar{P}_s \\
& \underline{E}_s \leq E_s(k) \leq \bar{E}_s
\end{aligned} \tag{4}$$

with actual dispatch $P_g(k) \in \mathbb{R}$, decision vector $X(k) = (P_g(k), E_s(k+1), P_s(k), P_s^+(k), P_s^-(k))^T$ and parameters $\tilde{P}_g(k), P_l(k), E_s^k, \underline{P}_s, \bar{P}_s, \underline{E}_s, \bar{E}_s$. Note, the state of energy in $k \in \mathbb{N}$ is known.

2.2 Forecast Evaluation Metrics for the Dispatchable Feeder

To assess forecasts with different characteristics, we define forecast evaluation metrics that measure forecast quality and forecast value in the context of the dispatchable feeder.

Forecast Quality. To evaluate the quality of the forecasts, we calculate the mean absolute error (MAE). For a given time series of observations y and predictions $\hat{y}$, the MAE is defined as

$$MAE = \frac{1}{N} \sum_{t=1}^N |\hat{y}_t - y_t|, \tag{5}$$

with sample size $N \in \mathbb{N}$. The MAE is a suitable measure for forecast quality since it is independent of the dispatchable feeder. A low MAE indicates that the predictions are, on average, similar to the actual observations and, therefore, indicates a high forecast quality.

Forecast Value. In order to evaluate the value of the forecasts for the dispatchable feeder, we consider the resulting costs from the dispatchable feeder optimisation problem. More precisely, we consider three average daily costs, namely, the average daily dispatch schedule costs, the average daily imbalance costs, and the average daily total costs. The dispatchable feeder aims to minimise costs, and therefore, lower costs imply a higher forecast value. We describe each of these costs used to measure forecast value in the following.

The first forecast value metric is the costs associated with the DS. For this metric, we use the costs in Equation (2). The second forecast value metric is the imbalance costs which are calculated after the execution of the optimisation problem. These costs result from the difference between the actual dispatch and the day-ahead DS. The imbalance costs are defined as

$$C_{imb}(\Delta P_g(k)) = c_{quadratic}^\Delta \cdot |\Delta P_g(k) \cdot \Delta t|^2 + c_{linear}^\Delta \cdot |\Delta P_g(k) \cdot \Delta t|$$

where $\Delta P_g(k)$ is the difference between the actual dispatch and the DS and $c_{quadratic}^\Delta \in \mathbb{R}_{\geq 0}$ and $c_{linear}^\Delta \in \mathbb{R}_{\geq 0}$ are weighting parameters¹. The total costs consider both of these aspects and are given as

$$C_{total}(\tilde{P}_g^+(k), \tilde{P}_g^-(k), \Delta P_g(k)) = C_{DS}(\tilde{P}_g^+(k), \tilde{P}_g^-(k)) + \alpha \cdot C_{imb}(\Delta P_g(k)),$$

with imbalance costs factor² $\alpha \in \mathbb{R}_{\geq 0}$.

To compare these three metrics for the forecast value, we calculate average daily costs. To calculate the average daily costs, we first aggregate the costs for each day. We then calculate the average daily costs by averaging these costs over all considered days. The result is the average daily costs for each of the considered metrics, i.e. the DS costs, the imbalance costs, and the total costs³.

The scale of the average daily costs strongly depends on the building's prosumption. Therefore, to compare the different forecasts associated with different characteristics, we assign a win to the forecast if it results in a building's minimal cost. Additionally, to analyse how clear the win is we consider the 5% win range. Thereby, a forecast results in an occurrence within the 5% win range if the percentage difference⁴ between the building's cost based on the forecast and the building's minimal cost over all forecasts is smaller or equal than 5%.

3 EVALUATION

In this section, we analyse the previously described forecast value for the dispatchable feeder. Therefore, Section 3.1 describes the experimental setup, while Section 3.2 present the results.

3.1 Experimental Setup

We briefly describe the experimental setup used for our evaluation. First, to report how we create forecasts with different characteristics, we explain the used forecast models. Further, we describe the data and hard- and software⁵.

Forecast Models. In order to create forecasts with different characteristics we use a simple neural network with varying loss functions. Therefore, we use fully connected neural networks with configuration shown in Table 1. These neural networks receive the historical prosumption of the past 24 hours as features. Based on this feature vector, the network predicts the next 42 hours of the prosumption. Further, in training, we use 20% of the training data to apply early stopping. As further hyperparameters we use a batch size of 512 and the RMSProb optimiser.

In order to create different forecast characteristics, we use seven different loss functions. A loss function is a function within the neural network that measures the neural network's ability to model the true values y . In training, the aim is to minimise the loss function. The first loss function is the **mean absolute error** (MAE) as defined in Equation (5). Since it is based on the absolute error, it treats

¹In this paper, we select $c_{quadratic}^\Delta = 0.05 \frac{\text{€}}{\text{kWh}^2}$ and $c_{linear}^\Delta = 0.3 \frac{\text{€}}{\text{kWh}}$ as in [2].

²In this paper, we select as imbalance costs factor either 2 or 10 as in [2].

³In the following, we use the average daily total costs without its unit (€).

⁴The percentage difference between costs C and minimal costs C_{min} is defined as $\frac{2 \cdot |C - C_{min}|}{|C| + |C_{min}|} \cdot 100$.

⁵Code to replicate the results is publicly available on <https://github.com/KIT-IAI/Impact-of-Forecast-Characteristics-on-Forecast-Value-for-Dispatchable-Feeder>.

Table 1: The architecture of the fully connected forecasting neural network

Layer	Activation function	Layer size
Input	Linear	24
Hidden	ReLU[7]	16
Output	Linear	42

overestimation the same as underestimation. Furthermore, larger errors are weighted the same as smaller errors. Second, the **mean squared error** (MSE) is the mean over the sum of the squared errors and defined as

$$\text{MSE} = \frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2.$$

Underestimation and overestimation are treated the same, analogous to the MAE. However, the quadratic term in the MSE implies that large errors, such as those caused by peaks, implicitly receive a higher weighting than with the MAE. Third, the **Huber** loss function combines properties of the MSE and the MAE and is defined as

$$\text{Huber} = \frac{1}{N} \sum_{t=1}^N \begin{cases} \frac{1}{2} (y_t - \hat{y}_t)^2, & \text{for } |y_t - \hat{y}_t| \leq 1 \\ (|y_t - \hat{y}_t| - \frac{1}{2}), & \text{otherwise} \end{cases}.$$

Therefore, if the absolute of the error is less than one, the squared error loss is used, and if not an absolute error-based loss is used. The last four loss functions are variations of the **pinball** loss. The pinball loss generates biased forecasts by underestimating the true values with probability of $\tau \in (0, 1)$ and overestimating the true values with probability of $\tau - 1$. This loss function is typically used to generate a τ quantile forecast and is defined as

$$\text{pinball}(\tau) = \frac{1}{N} \sum_{t=1}^N \max(\tau \cdot (y_t - \hat{y}_t), (\tau - 1) \cdot (y_t - \hat{y}_t)).$$

In the present paper, we consider the pinball losses for τ values of 0.1, 0.25, 0.75, and 0.9.

Data. The "Ausgrid - Solar home electricity data" set [10] is used to perform the evaluation on real-world residential load and PV power generation data. It includes data from 300 residential buildings in the Ausgrid Australia network area over the period from 1st July 2010 to 30th June 2013 in a 30-minute resolution. For our purpose, the data is resampled to hourly resolution by calculating the mean. In order to obtain varying data, we multiply the load $P_{\text{load}} \in \mathbb{R}_{\geq 0}$ and the PV power generation $P_{\text{PV}} \in \mathbb{R}_{\geq 0}$ with varying factors $\beta_{\text{load}}, \beta_{\text{PV}} \in \mathbb{R}_{\geq 0}$. The presumption is then

$$P_I(k) = \beta_{\text{load}} \cdot P_{\text{load}}(k) - \beta_{\text{PV}} \cdot P_{\text{PV}}(k).$$

Thereby, we consider the data **load5** with factors $\beta_{\text{load}} = 5, \beta_{\text{PV}} = 1$, **load1/2** with factors $\beta_{\text{load}} = 1/2, \beta_{\text{PV}} = 1$, **original** with factors $\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$, and **PV10** with factors $\beta_{\text{load}} = 1, \beta_{\text{PV}} = 10$.

Further, we split the data into a training subset and a testing subset. Therefore, the first two years are used as training subset. The last year is used as testing subset to calculate the forecast evaluation metrics for the dispatchable feeder in Section 2.2.

Table 2: The specification of the hardware used for the evaluation.

CPU Cores	32
CPU Clock Rate	2.1GHz
RAM	62 GB
GPU	Nvidia Titan RTX

Hard- and Software. All of our experiments are performed on a server with hardware specification provided in Table 2. To ensure reproducibility and reusability, we implement the experiments using the pyWATTS framework [8].

To solve the optimisation problem, we use the Python version of CasADi [1] with IPOPT [12]. The parameters of the optimisation problem are specified in Table 3. Thereby, to analyse the influence of the forecast characteristics on the forecast value depending on the battery capacity, we consider varying battery capacities $\bar{E}_s$ of 3, 6, and 13.5 kWh.

Table 3: The parameter specification of the optimisation problems in Equation (3) and Equation (4).

Parameter	Value
Δt	1 (hour)
$\mathcal{K}$	$\{k_s, \dots, k_s + 29\}^6$
$c_{\text{quadratic}}^+$	0.05 (€/kWh ²)
c_{linear}^+	0.3 (€/kWh)
$c_{\text{quadratic}}^-$	0.05 (€/kWh ²)
c_{linear}^-	0.15 (€/kWh)
$\bar{P}_s$	-5 (kW)
$\underline{P}_s$	5 (kW)
$\bar{E}_s$	0 (kWh)
$\underline{E}_s$	$\in \{3, 6, 13.5\}$ (kWh)
μ	0.05
E_s^0	day 1: $\frac{\bar{E}_s}{2}$ (kWh) days 2 - 7: estimated ⁷

3.2 Results

In the following, we evaluate and compare the forecasts with different characteristics regarding the forecast value for the dispatchable feeder on the testing subset. As we obtain the forecasts with different characteristics by using neural networks with different loss functions, we present the results of the forecasts by using the corresponding loss function's name. Thereby, we focus on three aspects. First, we analyse the relation between the considered forecast quality and the forecast value. Second, we study the forecast value for varying data. Finally, we examine the forecast value depending on varying battery capacities.

⁶ $k_s \in \mathbb{N}$ is the index of the time interval starting at midnight

⁷The initial state of energy E_s^0 for days two to seven is estimated via an optimisation problem, see [3].

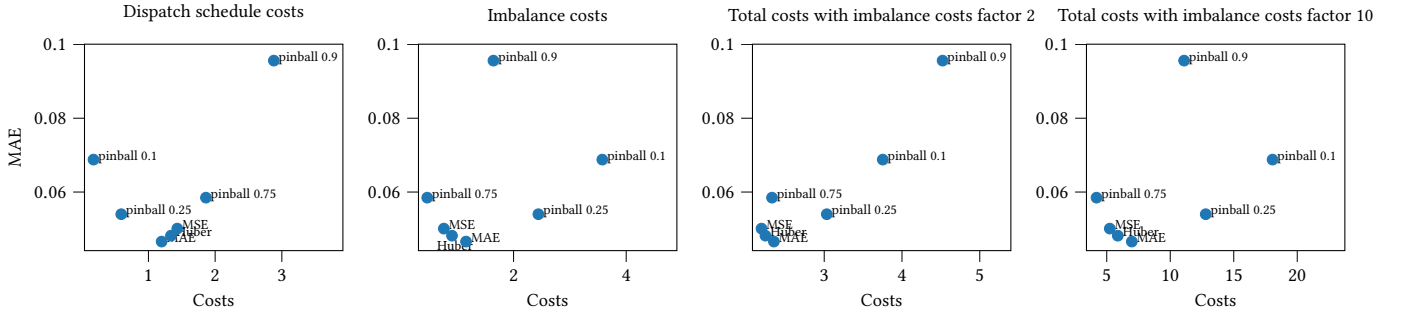


Figure 2: The MAE against the average daily dispatch schedule costs (left), the average daily imbalance costs (second to left), the average daily total costs with imbalance costs factor $\alpha = 2$ (second to right) and the average daily total costs with imbalance costs factor $\alpha = 10$ (right) for the considered loss functions. For this plot, we use the original data ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$) and battery capacity $\bar{E}_s = 13.5$ kWh. Note, for both the average daily costs and the MAE lower values are better.

3.2.1 Relation between Forecast Quality and Forecast Value. To investigate the relation between forecast quality and forecast value, we compare the MAE as forecast quality with the considered average daily costs as forecast values for the original data and a battery capacity of 13.5 kWh in a scatter plot (see Figure 2).

For all considered costs, we see that the forecasts resulting in the best forecast quality do not result in the best forecast value. Additionally, we observe that the relation between the forecast quality and the forecast value is not directly deducible. For the DS costs, the best and second-best forecast value is achieved with forecasts based on pinball 0.1 and pinball 0.25 respectively, although these have the second and fourth worst forecast quality, respectively. For the imbalance costs, we observe that the best forecast value is achieved for forecasts based on pinball 0.75, which have the third worst forecast quality. Considering the total costs with imbalance costs factor 2, the best forecast value is achieved with forecasts based on MSE, which only have the third-best forecast quality. Furthermore, the second-best forecast value is achieved with forecasts based on the Huber loss function, which have the second-best forecast quality, and the third best forecast value with forecasts based on MAE, which have the best forecast quality. When we consider the total costs with imbalance costs factor 10, as with the imbalance costs, the forecasts based on pinball 0.75 result in the best forecast value.

3.2.2 Comparison of different Loss Functions regarding varying Data. To compare the loss functions' forecast value regarding varying data, we analyse the number of wins and the occurrences within the 5% win range of each loss function for the data load5, load1/2, original, and PV10. Figure 3 shows the results for all considered average daily costs with a battery capacity of 13.5 kWh.

For the DS costs, pinball 0.1 achieves all 300 wins for all data except for PV10. Nevertheless, for PV10, pinball 0.1 achieves the most wins (133), followed by MSE (61), pinball 0.25 (38), and pinball 0.75 (37). With regard to the clarity of the wins, we observe that for load5, original, and load1/2 no other loss function except pinball 0.1 is within the 5% win range.

For the imbalance costs, we observe that the loss function resulting in the most wins depends on the data. For load5, MSE (137) leads to the most wins followed by Huber (71), pinball 0.75 (49), and MAE (40). In contrast, pinball 0.1 (0), pinball 0.25 (3) and pinball 0.9

(0) result in little to no wins. For original and load1/2, we observe a shift of the wins to the loss functions pinball 0.75 (213) and pinball 0.9 (61). For PV10, MAE (179) achieves the most wins, followed by Huber (69) and MSE (27). With regard to the clarity of the wins, we observe that for original and load1/2 the number of wins and the occurrences within the 5% win range are similar for all losses, indicating clear wins. In contrast, for load5 and PV10, the occurrences within the 5% win range are clearly higher than the actual wins.

Regarding the total costs with imbalance costs factor 2, we see that the winning loss functions are more heterogeneous compared to the DS or imbalance costs. For load5, we observe that Huber (109) and MAE (109) achieve the most wins followed by MSE (49). For original, pinball 0.75 (146) and MSE (111) have the most wins. With respect to the load1/2, the results are dominated by pinball 0.75 (242). A similar dominance is obtained with MAE (187) for PV10. In contrast to the DS and imbalance costs, we also see that the occurrences within the 5% win range are clearly higher than the number of wins for all data.

3.2.3 Comparison of different Loss Functions regarding varying Battery Capacities. In the last evaluation, we compare the loss functions with respect to the forecast value for varying battery capacities. Thereby, we calculate the wins and the occurrences within the 5% win range for the battery capacities 3 kWh, 6 kWh, and 13.5 kWh using the original data. Figure 4 shows the results for all average daily costs.

For the DS costs, pinball 0.1 achieves all wins regardless of the battery capacity. Thereby, no other loss function is within the 5% win range.

With respect to the imbalance costs, we observe for battery capacity 3 kWh that MSE (137) reaches the most wins followed by Huber (75), MAE (57) and pinball 0.75 (27). These are also the loss functions that are within the 5% win range. For battery capacity 6 kWh, we observe a shift of wins from Huber, MAE, and pinball 0.25 to pinball 0.75 (127) which reaches the most wins. For battery capacity 13.5 kWh, pinball 0.75 (213) achieves by far the most wins. The second most wins are reached by pinball 0.9 (61), followed by MSE (21) and Huber (4). Thereby, the occurrences within the 5% win ranges are only slightly higher than the number of wins for each loss function.

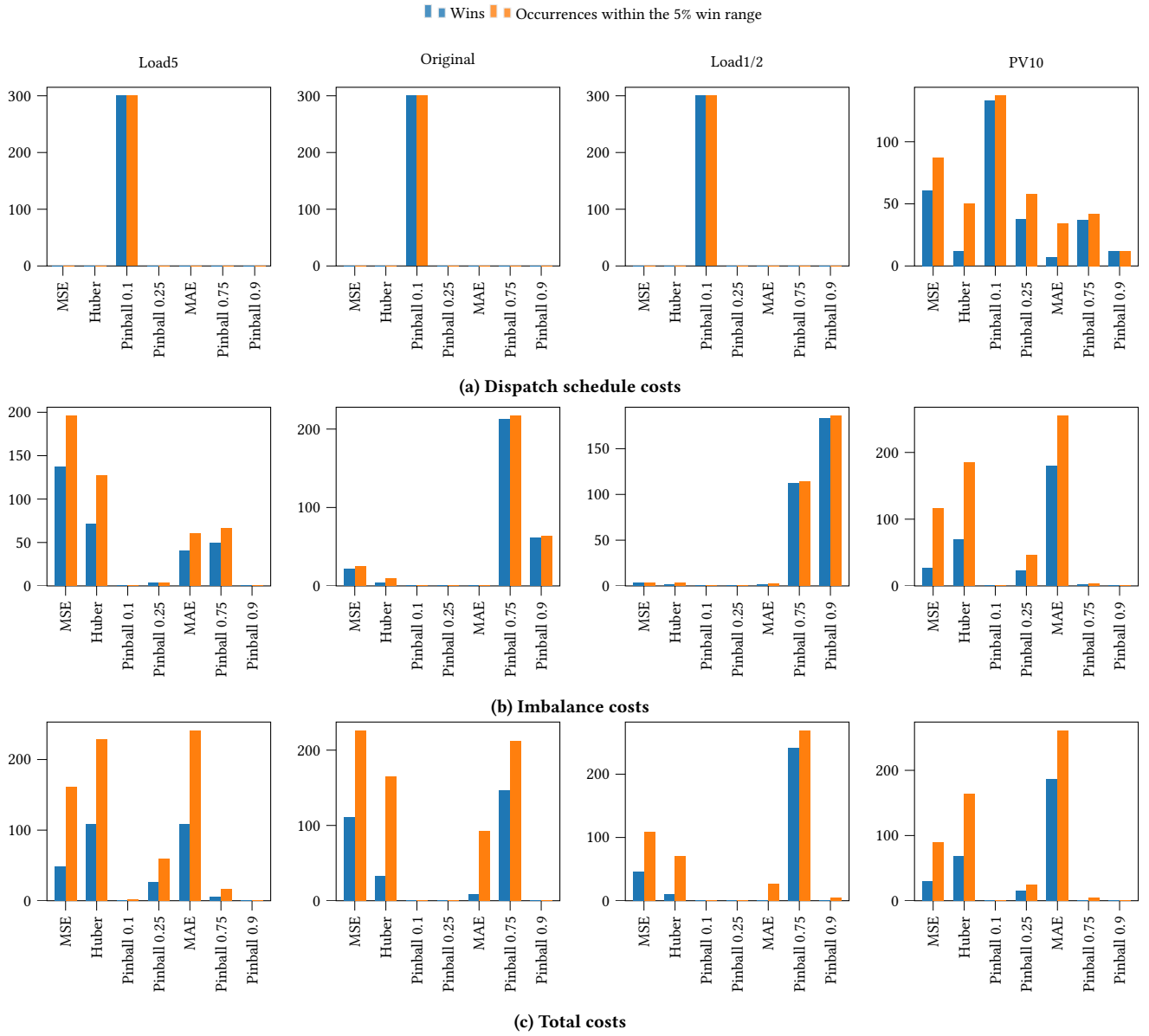


Figure 3: The wins and the occurrences within the 5% win range with respect to the average daily dispatch schedule costs (a), the average daily imbalance costs (b), the average daily total costs with imbalance costs factor $\alpha = 2$ (c) for the considered loss functions. Further, we consider the data load5 ($\beta_{\text{load}} = 5, \beta_{\text{PV}} = 1$) (left), original ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$) (second to left), load1/2 ($\beta_{\text{load}} = 1/2, \beta_{\text{PV}} = 1$) (second to right), and PV10 ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 10$) (right) as well as the battery capacity $\bar{E}_s = 13.5$ kWh. Note, higher win values and higher occurrences within the 5% win range are better.

For the total costs with imbalance factor 2, we observe a noticeable difference between a loss function's wins and a loss function's occurrences within the 5% win range for all battery capacities. For battery capacity 3 kWh, we see that MAE (153) reaches the most wins followed by Huber (74), pinball 0.25 (37), and MSE (35). Thereby, the occurrences within the 5% range are similar for MAE (254) and Huber (211). For battery capacity 6 kWh, MSE (130)

achieves for the most buildings the minimal costs followed by Huber (85) and MAE (56). With respect to the occurrence within the 5% win range, MSE (247) and Huber (252) lead to comparable results. For battery capacity 13.5 kWh, the most wins are reached with pinball 0.75 (146), while the occurrences within the 5% win range are similar for pinball 0.75 (212) and MSE (226).

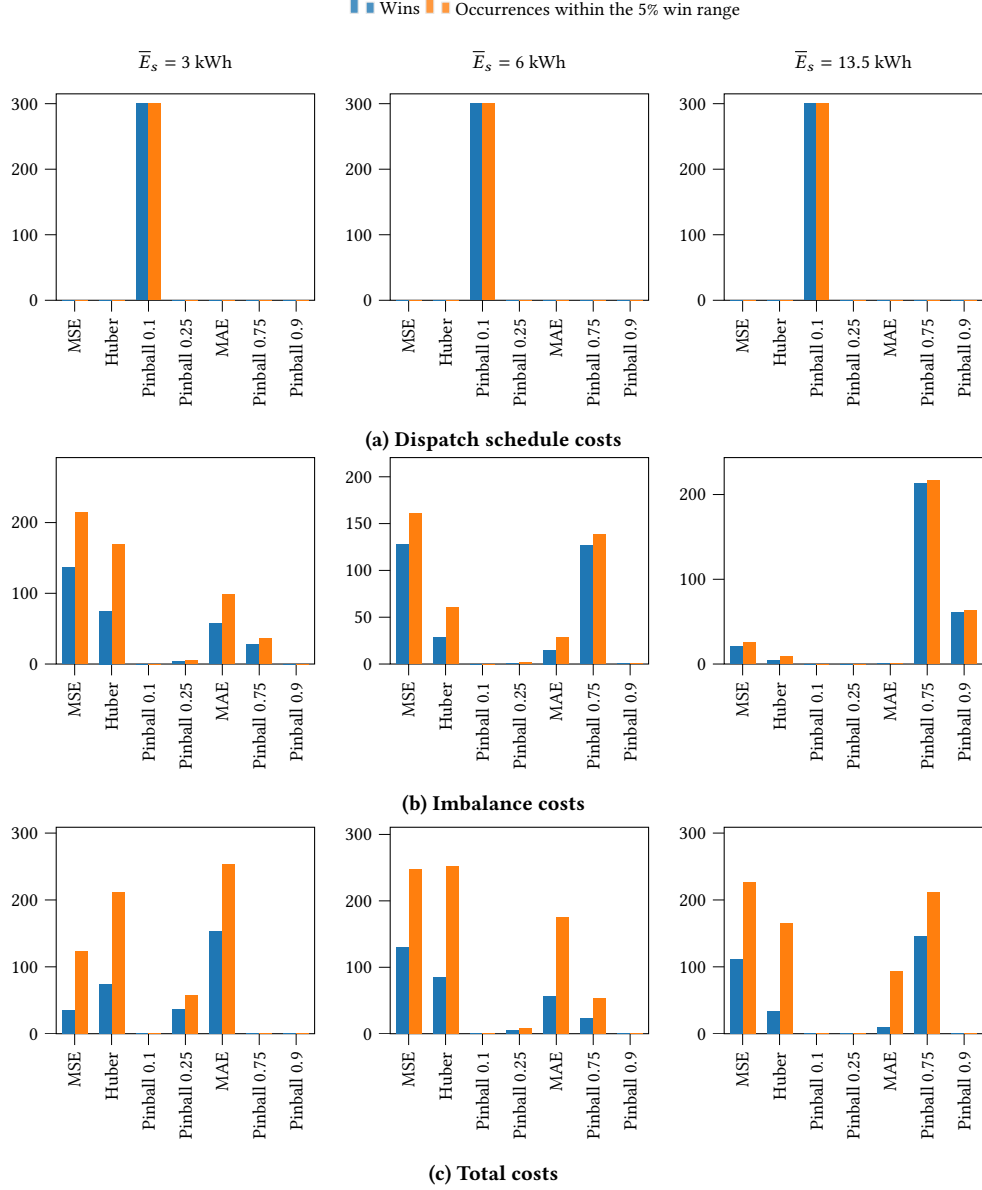


Figure 4: The wins and the occurrences within the 5% win range with respect to the average daily dispatch schedule costs (a), the average daily imbalance costs (b), the average daily total costs with imbalance costs factor $\alpha = 2$ (c) for the considered loss functions. Further, we consider battery capacities $\bar{E}_s = 3$ kWh (left), $\bar{E}_s = 6$ kWh (center), and $\bar{E}_s = 13.5$ kWh (right) as well as the original data ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$). Note, higher win values and higher occurrences within the 5% win range are better.

4 DISCUSSION

In this section, we discuss the previously reported results in Section 3.2. Thereby, we focus on the following two aspects. First, we discuss the relation between forecast quality and forecast value. Second, we discuss the influence of the forecast characteristics on the forecast value.

4.1 Relation between Forecast Quality and Forecast Value

In Section 3.2.1, the results show that the relation between the considered forecast quality, namely the MAE, and the considered forecast values for the dispatchable feeder, namely the average daily costs, is non-monotonic. This means that improving the forecast quality does not necessarily improve the forecast value for the dispatchable feeder and thus can result in higher average daily costs. This result indicates that it can be beneficial for the performance

of the dispatchable feeder to consider not only the forecast quality but also the forecast value when creating the required forecast.

4.2 Influence of Forecast Characteristics

Regarding the forecast characteristics' influence on the forecast value, we focus on two major aspects. First, the results in Section 3.2.1 as well as the results in Section 3.2.2 and Section 3.2.3 demonstrate that the forecast characteristics affect highly the considered forecast value for the dispatchable feeder. When aiming for the best forecast value one might want to look beyond the usual forecast characteristics determined by standard loss functions. Second, how the forecast characteristics affect the considered forecast value depends on the data and the battery capacity. Furthermore, we observe that this impact varies for each forecast value. Thus, we further discuss this impact from the results described in Section 3.2.2 and Section 3.2.3 for each considered forecast value, namely the average daily costs, separately.

Influence on Dispatch Schedule Costs. For the DS costs, the results indicate that underestimating the prosumption leads by far to the lowest costs for all battery capacities and data except for PV10. A possible reason why an underestimation of the prosumption is beneficial in most cases is that this leads to an underestimation of the load and an overestimation of the PV power generation. Thus, hardly any power is imported from the grid according to the resulting DS, see Figure 5 (b). As consequence, the costs associated with the DS are small. However, when the PV power generation is scaled with 10, a strong underestimation becomes disadvantageous. A possible reason, therefore, is that the building generates already so much PV power that an underestimation of the prosumption and thus an overestimation of the PV power generation can not be consumed by the building. Consequently, more PV power is fed into the grid which raises costs, see Figure 5 (a).

Influence on Imbalance Costs. With respect to the imbalance costs, we observe that the preferred forecast characteristics depend on the data and the battery capacity. We observe that overestimation becomes more beneficial the smaller the load factor and the bigger the battery capacity indicating asymmetric imbalance costs. In the following, we aim to provide a possible explanation of this behaviour. If the prosumption is overestimated, the DS is calculated assuming that the building consumes more and generates less energy than it actually does. Thus, in the second level, the battery then charges if possible to mitigate imbalances. As a result, the battery has energy reserves for unexpected high prosumption. This behaviour is beneficial if the battery is not fully charged and thus can still balance out unexpected low prosumption. For a higher load/PV power generation factor or a smaller battery, the battery reaches faster its limit. As a result, imbalances become more likely, see Figure 6.

Influence on Total Costs. As the total costs are the weighted sum of the DS costs and the imbalance costs, the performance of the forecast characteristics with respect to the two latter costs impacts the performance of the forecast characteristics with respect to the total costs. Thereby, the imbalance costs factor determines the impact of each cost. For both imbalance costs factors, we observe that an overestimation can be beneficial depending on the data and

the battery capacity, e.g. Figure 2. More specifically, for imbalance costs factor 10, the imbalance costs make up the largest share of the total costs. As consequence, we observe the same results as for the imbalance costs, see Figure 7. On the other hand, for an imbalance costs factor of 2, the impact of the imbalance costs is less dominant. Therefore, the impact of the DS costs becomes more important and the results appear to be a combination of the results for both costs, see Figure 3 and Figure 4.

5 CONCLUSION

The present paper analyses the forecast value for the dispatchable feeder. More precisely, we show that the relation between the considered forecast quality and forecast value is non-monotonic. Further, we demonstrate that the characteristics of the forecast influence the forecast value and that this influence depends on problem-specific parameters such as the data and the battery capacity. The results of the present paper indicate that besides the forecast quality, one may want to consider the forecast value when creating forecasts for the dispatchable feeder. Thus, to improve the performance of the dispatchable feeder, it is beneficial to carefully consider the forecast characteristics, taking into account the considered forecast value, the data, and the battery capacity.

In future work, we aim to develop a framework that automatically selects the forecast leading to the highest forecast value for a dispatchable feeder, taking into account the building-specific data and the battery capacity. Further, since the used forecast models are rather simple, we want to include more advanced forecast models in further analyses. Additionally, as the observations are only valid for the dispatchable feeder, we want to extend the analysis to a more general setting.

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A INFLUENCE OF UNDERESTIMATION ON DISPATCH SCHEDULE COSTS

To demonstrate the influence of underestimation on the dispatch schedule costs, Figure 6 shows an exemplary week's results of using pinball 0.1 for the data PV10 and original. For original, we observe that the absolute value of the DS stays under 0.6 kW resulting in small dispatch schedule costs. In contrast, for PV10, the DS reaches values up to -12 kW which raises the costs.

B INFLUENCE OF OVERESTIMATION ON IMBALANCE COSTS

To demonstrate the influence of overestimation on the imbalance costs, Figure 6 shows an exemplary week's results of using pinball

0.75 for the data load5 and original. For both data, the actual battery's state of energy is often higher than the scheduled battery's state of energy (SoE) due to the overestimation of the prosumption. Further, for load5, we observe that the battery is often fully charged. As consequence, the battery can not balance out the forecast errors and imbalances occur. In contrast, for original, the battery has reserve for forecast errors and only little imbalances occur.

C INFLUENCE OF FORECAST CHARACTERISTICS ON TOTAL COSTS WITH IMBALANCE COSTS FACTOR 10

Figure 7 shows the wins and the occurrences within the 5% win range with respect to the total costs with imbalance costs factor 10 for varying data and varying battery capacities. The results are similar to the results with respect to the imbalance costs.

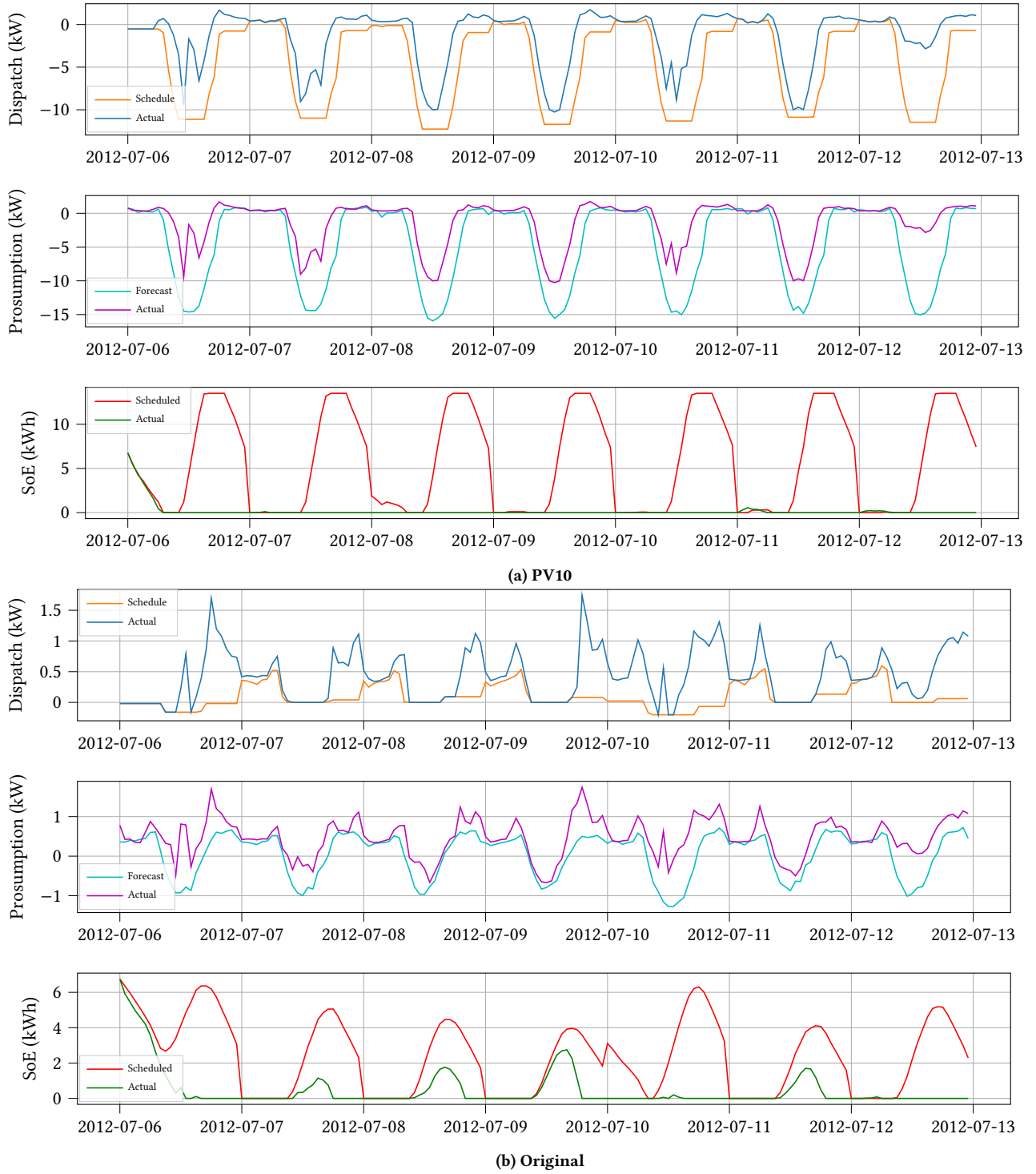


Figure 5: The dispatch schedule $\tilde{P}_g$, the actual dispatch P_g , the prosumption forecast $\hat{P}_l$, the actual prosumption P_l , the scheduled battery's state of energy and the actual battery's state of energy resulting from the usage of pinball 0.1 for the data PV10 ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 10$) (a) and original ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$) (b) for an exemplary week.

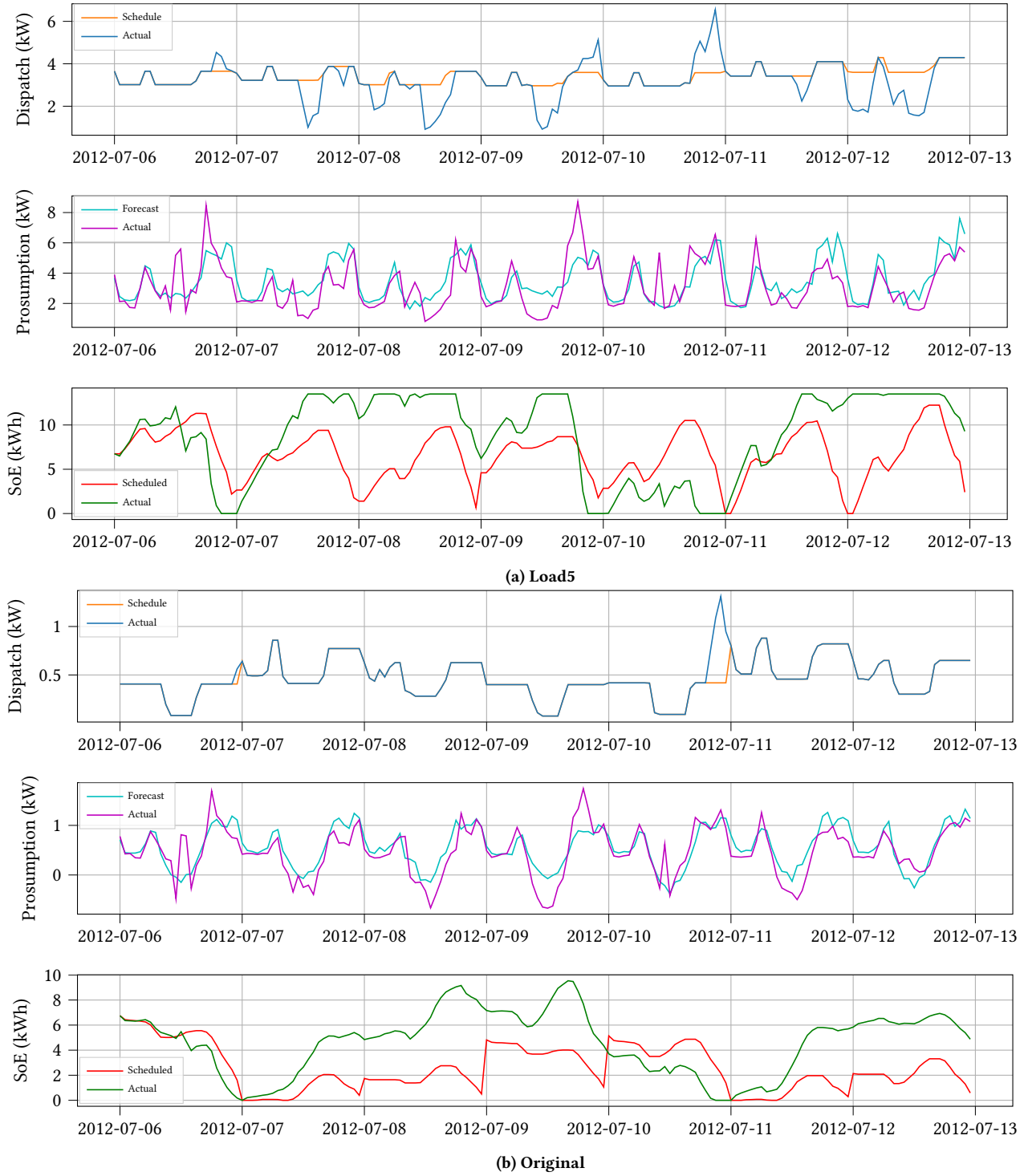


Figure 6: The dispatch schedule $\tilde{P}_g$, the actual dispatch P_g , the prosumption forecast $\hat{P}_l$, the actual prosumption P_l , the scheduled battery's state of energy and the actual battery's state of energy resulting from the usage of pinball 0.75 for the data load5 ($\beta_{\text{load}} = 5, \beta_{\text{PV}} = 1$) (a) and original ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$) (b) for an exemplary week.

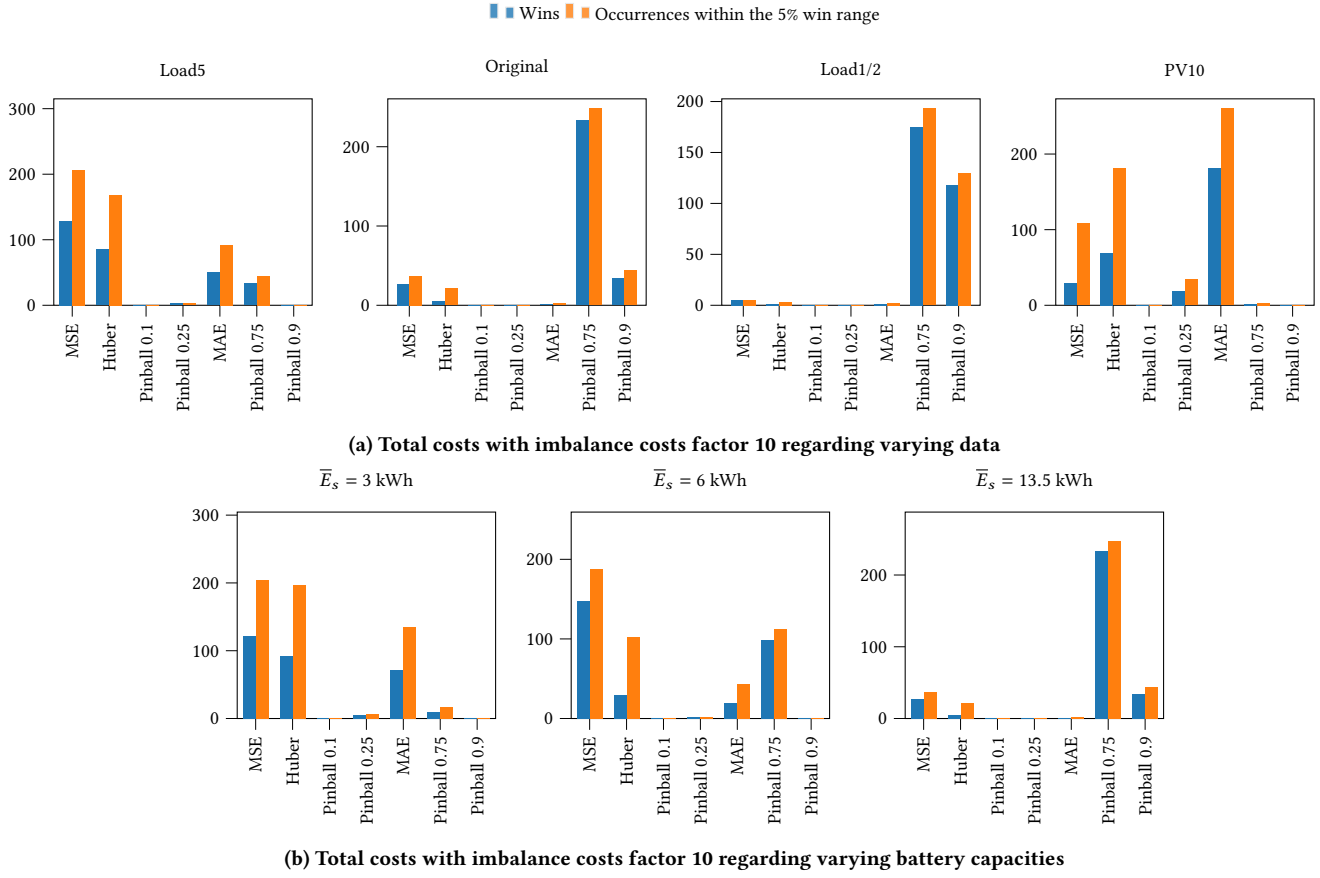


Figure 7: The wins and the occurrences within the 5% win range with respect to the average daily total costs with imbalance costs factor $\alpha = 10$ for the considered loss functions. Further, in (a), we consider the data load5 ($\beta_{\text{load}} = 5, \beta_{\text{PV}} = 1$) (left), original ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$) (second to left), load1/2 ($\beta_{\text{load}} = 1/2, \beta_{\text{PV}} = 1$) (second to right), and PV10 ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 10$) (right) as well as a battery capacity $\bar{E}_s = 13.5$ kWh. In (b), we consider battery capacities $\bar{E}_s = 3$ kWh (left), $\bar{E}_s = 6$ kWh (center), and $\bar{E}_s = 13.5$ kWh (right) as well as the original data ($\beta_{\text{load}} = 1, \beta_{\text{PV}} = 1$). Note, higher win values are better.